Robust Overlap Coherence of the Reset Packet Atlas

Principal Angles, Polar Transitions, and a Finite Conditioning Wall

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Abstract

RH-151 certified an independent clock-rank top-packet ball at each of 130 source-memory snapshots. Spectral existence alone does not produce a moving frame: consecutive exact packet spaces must retain a full-rank overlap after both endpoint balls are inserted.

Let U, V be nominal orthonormal frames, let $\alpha = \sigma_{\min}(U^*V)$, and suppose exact projectors lie within radii ϵ_U, ϵ_V. We prove two outward overlap lowers. The principal-angle triangle gives

$$\widetilde{\alpha} \geq \cos(\arccos \alpha + arcsine\epsilon_U + arcsine\epsilon_V)_+ ,$$

while polar-aligned frames give

$$\widetilde{\alpha} \geq (\alpha - f(\epsilon_U) - f(\epsilon_V))_+ , \quad f(e) = \sqrt{2 - 2\sqrt{1 - e^2}}.$$

Their maximum is a valid robust lower. A positive lower controls both the inverse overlap and the canonical polar transition; the zero boundary is sharp because the packet balls can then contain orthogonal subspaces.

All 120 frozen consecutive transitions certify as invertible. The minimum robust overlap is 8.9866×10^{-5}, the maximum inverse-overlap upper is 1.1128×10^4, and every polar transition is stable with radius below 0.01170. Thus the reset route remains open at finite horizon, but not with a uniform conditioning constant: 19 transitions are below 0.1, six below 0.01, and two below 0.001. The next layer must propagate these actual condition numbers through the outward Gram/tail assembly.

1 The overlap interface

For equal-rank orthogonal projectors P, Q, the largest principal angle $\theta_{\max}$ satisfies

$$\|P - Q\| = \sin \theta_{\max}, \qquad \sigma_{\min}(U^*V) = \cos \theta_{\max}$$

for any orthonormal frames U, V of their ranges. A transition map is invertible exactly when $\theta_{\max} < \pi/2$.

RH-151 supplies nominal frames U_t and projector balls $\|\widetilde{P}_t - P_t\| \leq \epsilon_t$. The aim is an outward lower for $\sigma_{\min}(\widetilde{U}^*_{t-1}\widetilde{U}_t)$ independent of frame signs and internal rotations.

2 Two robust overlap lowers

The largest principal angle is a metric on the Grassmannian [Stewart and Sun, 1990]. Therefore

$$\widetilde{\theta} \leq \arcsin \epsilon_U + \arccos \alpha + \arcsin \epsilon_V .$$

Theorem 1 (Angular overlap lower)**.** *With the notation above, put*

$$a = \arccos \alpha + \arcsin \epsilon_U + \arcsin \epsilon_V .$$

If $a < \pi/2$, then every pair of exact packet spaces in the two balls obeys

$$\sigma_{\min}(\widetilde{U}^* \widetilde{V}) \geq \cos a.$$

If $a \geq \pi/2$, no positive lower follows from the three scalar data alone.

Proof. The triangle inequality proves the first statement. At equality, rank-one lines in a common plane can spend the two ball angles in opposite directions until the exact lines are orthogonal, proving the information boundary. □

Projector balls also give canonically aligned frames. If $\|\widetilde{P} - P\| \leq e$, polar alignment produces frames satisfying

$$\|\widetilde{U} - U\| \leq f(e) := \sqrt{2 - 2\sqrt{1 - e^2}}.$$

Theorem 2 (Frame overlap lower)**.** *After independent polar alignment at both endpoints,*

$$\|\widetilde{U}^* \widetilde{V} - U^* V\| \leq f(\epsilon_U) + f(\epsilon_V) =: \eta.$$

Consequently

$$\sigma_{\min}(\widetilde{U}^* \widetilde{V}) \geq (\alpha - \eta)_+ .$$

Proof. Expand the overlap difference into one left-frame and one right-frame term; both untouched frames have operator norm one. Singular-value Weyl bounds give the result [Bhatia, 1997]. □

We use the maximum of the angular and frame lowers. Neither dominates in all geometries, and retaining both costs nothing.

3 Inverse and polar transition control

Let $C = U^* V$ and let $R = C(C^* C)^{-1/2}$ be its unitary polar factor. A robust overlap lower $\underline{\alpha} > 0$ immediately gives

$$\|\widetilde{C}^{-1}\| \leq \underline{\alpha}^{-1}.$$

For the polar factors, a standard perturbation inequality gives

$$\|\widetilde{R} - R\| \leq \frac{2\|\widetilde{C} - C\|}{\sigma_{\min}(C) + \sigma_{\min}(\widetilde{C})} \leq \frac{2\eta}{2\alpha - \eta}$$

threshold	positive	$< 10^{-1}$	$< 10^{-2}$	$< 10^{-3}$	zero
transitions	120	19	6	2	0

Table 1: Robust reset-overlap counts.

when $\alpha > \eta$ [Li, 1995]. Thus overlap invertibility and canonical frame transport are certified by the same endpoint packet balls.

The inverse factor is the dangerous quantity. A transition can have a very stable polar rotation while its overlap is almost singular: the rotation depends on orientation, whereas inverse amplification depends on the weakest cosine.

4 The 120-transition audit

The audit rebuilds the ten RH-151 reset-frame chains and inserts the archived projector radii at both endpoints of every consecutive transition. Table 1 summarizes the robust overlap distribution.

All 120 overlap maps remain invertible. The median robust lower is 0.90183. The minimum, 8.9866×10^{-5}, occurs on the first $\sigma = 0.01$ right transition and gives inverse upper 1.1128×10^{4}. The largest polar-transition radius, 0.011695, occurs elsewhere, on the fine left chain. Hence the worst inverse and worst rotation uncertainty are distinct mechanisms.

The maximum cumulative chain drawdown

$$\sum_t -\log \underline{\alpha}_t$$

is 32.015. This is finite and far smaller than the direct projective products rejected in RH-146, but it is not evidence for a uniform asymptotic lower.

5 Consequence and boundary

The reset atlas is not merely a collection of isolated eigenspaces. It supports a finite, canonically aligned moving frame on all five scales and both channels. The finite route to an outward assembly therefore remains open.

What fails is uniform conditioning. An argument that replaces all overlap maps by one harmless constant would discard a factor above 10^4 at a single birth transition. RH-153 must derive Gram and tail transport with the actual transition-by-transition inverse bounds, then determine whether the large factors are multiplied, canceled, or confined to a finite prefix.

We have not built that outward assembly, proved a uniform overlap lower, closed the reset source-to-support interface, established Stage A, constructed a Hilbert–Polya operator, identified zeta zeros, or proved the Riemann Hypothesis.

References

Rajendra Bhatia. *Matrix Analysis*. Springer, 1997.

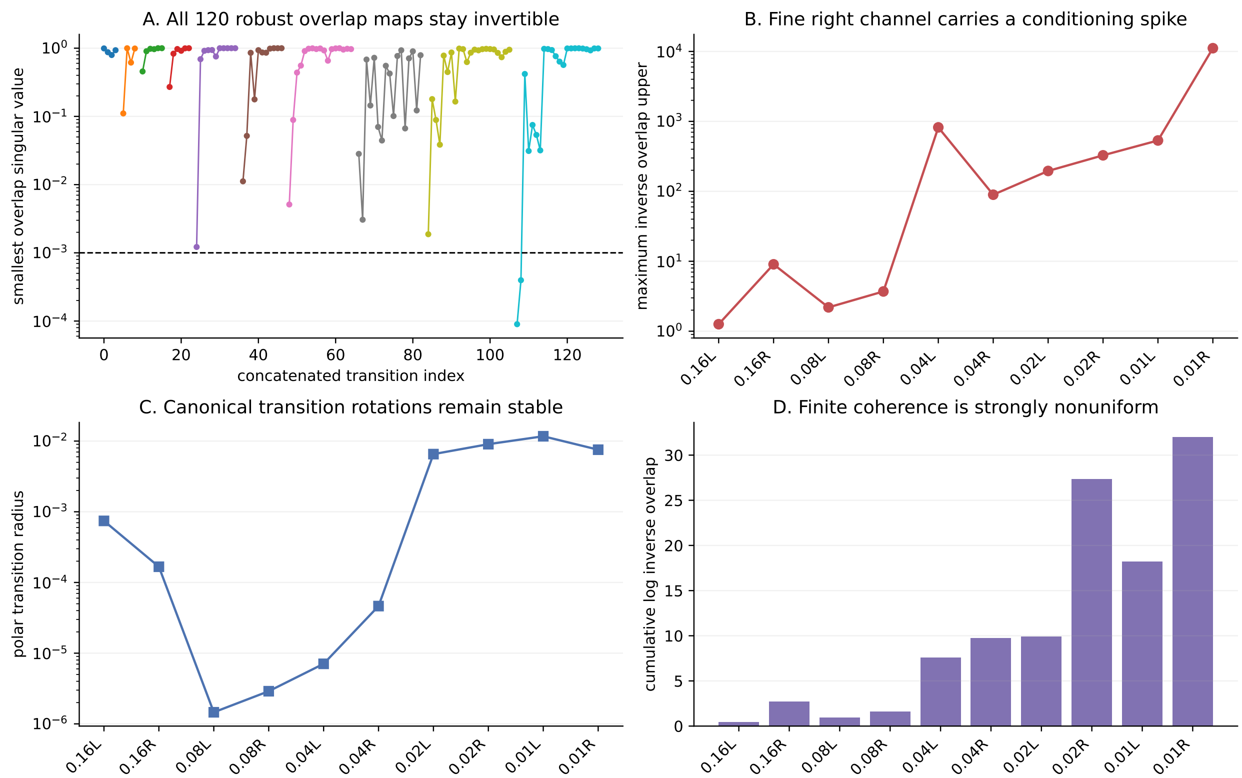

Figure 1: Robust overlaps, inverse amplification, polar radii, and cumulative conditioning drawdown across the ten reset chains.

Ren-Cang Li. New perturbation bounds for the unitary polar factor. *SIAM Journal on Matrix Analysis and Applications*, 16(1):327–332, 1995.

G. W. Stewart and Ji-Guang Sun. *Matrix Perturbation Theory*. Academic Press, 1990.